

Results: All patients showed the major response : Twenty-two (81.5%) patients achieved complete response, 5 patients (18.5%) achieved partial response at median F-U of 2.25 months after CKS. One year and 2 year relapse free survival rate were 78.8 % and 62.4%, respectively. However, during follow-up periods, grade 3 $\geq$ late complications were observed in 9 patients (33.3%) (2 pontine necrosis in NPC, 3 soft tissue necroses, unhealed mucosal ulcer with fatal bleeding in 1 tongue cancer, 1 optic neuropathy and 2 retinopathy in patients with GTV close to eye ball). And also grade 1~2 manageable late complication observed in 9 patients (minimal but prolonged mucosal ulcer over 1 year in one patients, 3 cataracts, 3 nasolacrimal duct stenosis or occlusion, soft tissue and skin fibrosis in one patients, and unhealed wound dehiscence in one patients).

Conclusions: Our results show that CKS can be an effective boost modality to improve the local control of locally advanced head and neck cancer through dose escalation for the high risk area, but late complication rate were unacceptably high. FSRT with large fractional dose should be applied with caution in head and neck sites. It would be needed more accumulated clinical data. Further details of our data and recommendation will be discussed.

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2588 The Importance of Patient Selection for the Postoperative Concurrent Chemoradiotherapy (CCRT) to Improve Therapeutic Gain for Locally Advanced Squamous Cell Carcinoma of the Head and Neck (SCCHN)

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Purpose/Objective(s): We retrospectively evaluated the role of postoperative CCRT in the patients with locally advanced SCCHN and focused on the importance of patients' selection to obtain therapeutic gain in trimodality therapy.

Materials/Methods: From March, 1993 to July, 2008, 112 patients with locally advanced SCCHN who had undergone macroscopically complete resection were evaluated. We compared the overall survival (OS) and the disease-free survival (DFS) between postoperative radiotherapy arm in early period (1993~2003) and postoperative CCRT arm in late period (2003~2008). The cumulative incidence of local or regional relapse, metastasis, secondary primary tumors and acute adverse effects were analyzed.

Results: The median follow-up period of the patients was 71 months. The locoregional failure rate for the radiotherapy arm was significantly higher than that for the CCRT arm (20.9% vs. 4.0%, $p = 0.01$). Even though, having of unfavorable prognostic factors in CCRT arm, there was a trend of improved OS and DFS at three years in favor of the CCRT arm (70% vs. 52.5%, $p = 0.772$, 66.1% vs. 58.5%, $p = 0.226$). However, there is no difference of survival between postoperative RT arm and CCRT arm within 1 year, then the survival gap appeared after 1 year. 13 patients were dead early during the first year of follow-up in CCRT arm. These early succumbs in CCRT arm had characteristics compared with remaining patients: poor performance status, extracapsular-extended node, old age over sixty ($p < 0.05$). The incidence of grade 3 or 4 hematologic and mucosal toxicity was significantly higher in the CCRT arm than that in the radiotherapy arm. (38.0% vs. 1.6%, $p = 0.001$, 40.0% vs. 19.4%, $p = 0.030$, respectively).

Conclusions: Our study suggests that postoperative CCRT is more efficacious than radiotherapy alone in patients with locally advanced SCCHN. However, trimodality therapy of curative surgery followed by chemoradiotherapy is too intensive and toxic for patients of poor performance status and old aged patients. Thus, physicians consider less intensive treatment option. For patients of high risk of early distant metastasis is also needed sophisticated treatment decision such as neoadjuvant chemotherapy or definitive CCRT to improve therapeutic gain.

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2589 Choroidal Melanoma Brachytherapy: An Effectiveness Comparison between Cobalt and Ruthenium

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Purpose/Objective(s): To evaluate and compare the preliminary results of tumor control and ocular preservation in patients with small or medium sized choroidal melanoma treated with ruthenium (Ru-106) or cobalt (Co-60) brachytherapy.

Materials/Methods: A retrospective comparative analysis of 97 out of 146 patients that underwent Co-60 brachytherapy between 1995 January and 2003 June and 83 out of 94 patients that underwent Ru-106 brachytherapy between 2004 January and 2008 September was performed. All patients were 18 age or older at the time of implant and were staged as T1/T2 non-metastatic melanoma. All of them were treated at the same single institution and had a minimal follow-up of 6 months. The main parameters evaluated were tumor control (Local Control - LC; Progression Free Survival - PFS) and ocular preservation (Enucleation Free Survival - EFS). LC, PFS, and EFS were calculated using Kaplan-Meier method and the group's comparison was performed using log-rank test.

Results: The median follow-up was 83 (12-165) months (Co-60) and 22 (6-59) months (Ru-106). Median age for each group was 55 (Co-60) and 60 (Ru-106). The median values of height and maximal basal diameter were comparable between the groups (4.2/9.7 mm for Co-60 and 4.3/9.3 mm for Ru-106) as were the median apical and basal doses: 108.8/265 Gy with Co-60 and 100.8/307.4 Gy with Ru-106. There were 22 local, 11 distant relapses and 31 enucleations in Co-60 group, and there were 6 local, 1 distant relapse and 6 enucleations in the other group. Actuarial 2 and 4 years LC, PFS and EFS were, respectively, for Co-60: 97/89%; 96/81%; 95/88%. For Ru-106, these numbers were: 91/84%; 91/84%; 93/87%. The comparison showed no statistic difference between the groups (LC: $p = 0.404$; PFS: $p = 0.834$; EFS: $p = 0.736$).

Conclusions: Small and medium sized choroidal melanomas can be adequately treated with Co-60 or Ru-106 brachytherapy, with high tumor control and ocular preservation rates. The two techniques are comparable and have equally outcomes. A longer follow-up may clinically show the dosimetric superiority of Ru-106 plaque.

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2590 Prospective Study to Determine Ideal Dose Constraints for the Oral Cavity in Patients Undergoing Intensity Modulated Radiotherapy for Head and Neck Cancer

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Purpose/Objective(s): Oral mucositis is the primary acute dose limiting factor in patients receiving radiotherapy for head and neck cancer. However, there are minimal data identifying the optimal dose constraints to be used for IMRT planning. This study was undertaken to prospectively evaluate oral mucositis during IMRT and correlate patient symptoms with dosimetric parameters to the oral cavity in attempt to better refine treatment-planning guidelines.

Materials/Methods: The study population consisted of 14 head and neck cancer patients, treated definitively with IMRT to a median dose of 70 Gy (range, 50.4Gy to 70 Gy). 7 patients received concurrent cisplatin-based chemotherapy. The oral cavity was contoured on all of the patients' planning CT scans, and dose-volume data was prospectively recorded. During treatment, each patient was examined at least once weekly and asked to complete a standardized symptom questionnaire designed to assess quality of life (QOL). Oral mucositis was graded according to the Common Terminology Criteria for Adverse Events (CTCAE) version 4 guidelines.

Results: The average mean and maximum doses to the oral cavity were 38.9 Gy (range, 3.0 to 54.1 Gy) and 62.12 Gy (range, 11.84 to 75.32 Gy). The mean V20, V30, V45, and V50 statistics were: 92.7% (range, 0-100%), 70.0% (range, 0-100%), 37.4 % (range, 0-74.4%), and 24.9% (range, 0-58.4%). The incidence of grade 0, 1, 2, and 3 oral mucositis was 7%, 14%, 64%, and 14%. None of the patients developed grade 4 or 5 toxicity. A statistically significant correlation was detected between increasing oral mucositis and both V45 and V50 ($p < 0.05$). This was significant for patients treated by both radiation therapy alone and those treated by chemoradiation ($p < 0.05$). A correlation was also observed between increasing mouth pain, and decreased QOL. The lowest QOL scores were observed in patients who developed grade 2 and 3 oral mucositis.

Conclusions: The severity of oral mucositis appears to correlate with dose-volume characteristics to the oral cavity. Our results suggest that aiming for a V45 less than 40% and a V50 less than 24% would limit acute toxicity related to mucositis and improve patients' quality of life during treatment.

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2591 Consequences of Pharyngeal Constrictor Contouring Method on Dose Calculation and Optimization

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Purpose/Objective(s): Dysphagia and aspiration are important sequella following chemoradiation for head and neck cancer. Recent reports detail dose-function relationships for swallowing structures including the pharyngeal constrictors. However, different methods exist to delineate the constrictors: 1) including the full lateral and anterior extent of the constrictors as they attach to pterygomandibular raphe, cartilages, hyoid bone, and pterygoid plates (anatomical pharyngeal constrictors, APC) and 2) including only the posterior pharyngeal wall (PPW). The different delineation methods may affect reported doses and dose-response relationships. This study aimed to examine consequences of this variability on optimization and dose reporting.

Materials/Methods: The PPW was contoured on contrast-enhanced planning CT datasets of 26 patients with advanced oropharyngeal cancer enrolled on a prospective study of chemo-IMRT aimed at sparing the swallowing structures, for which the APC were delineated for clinical purposes. Volumes and calculated dose received by the PPW and APC were compared in the clinical plans. Then, using a multi-criteria hierarchical optimization method, 15 cases were re-optimized by a single investigator, first using the APC and then using the PPW to determine whether there would be a difference in the degree of sparing achievable, and in the calculated dose to the constrictors.

Results: APCs were significantly larger than PPWs, with mean volumes $\pm$ standard deviations of 30 ± 9 and 16 ± 4 cc, respectively, $p < 0.001$. Averaging over the 26 patients, calculated clinical (APC-optimized) mean doses to APC were significantly higher than for PPWs, with mean dose $\pm$ standard deviation of 56 ± 9 and 51 ± 8 Gy, respectively, $p < 0.001$. The higher doses in APCs were due to larger overlap laterally with planning target volumes (PTVs) in the tonsil and high neck. 15 cases were re-optimized based on PPW volumes keeping target coverage identical to the APC-based optimization. Mean doses to the APC and PPW were 58 ± 8 and 52 ± 8 respectively (PPW-optimized), compared with 59 ± 7 and 54 ± 7 (APC-optimized). Overall, there was only a 1 ± 1 Gy mean difference for each constrictor structure between the PPW-based or APC -based optimizations.

Conclusions: There are substantial spatial and volumetric differences between APC and PPW volumes, and larger overlap of APC volumes with the PTVs. Although these differences do not generally affect the ability to spare these structures in the optimization process, since target coverage has priority over normal tissue sparing, they do affect the overall reporting of dose.

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2592 Treatment Outcomes of Different Chemotherapy Sequences in N3 Stage Nasopharyngeal Carcinoma

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Purpose/Objective(s): Although concurrent chemoradiotherapy has been proven standard of locoregionally advanced nasopharyngeal carcinoma (NPC), the risk of local relapse and distant metastasis is extremely high in patients with N3 disease. Our study is aim to compare different sequences of chemotherapy in these patients retrospectively.